Charles Shang

Principal ML Engineer - Generative Video, World Models, and 3D Generation

San Francisco, CA | linkedin.com/in/charlesshang | shangchong90@gmail.com | github.com/CharlesShang | charlesshang.github.io

PROFILE

Hands-on ML research engineer with 10+ years in computer vision and machine learning, specializing in 3D generation, generative video, and world models. Work directly across the full model development lifecycle: representation and tokenizer design, data pipelines, distributed training, evaluation, and production inference. Co-developed Roblox Cube and took Cube MeshGPT from research experiments to production for controllable 3D asset generation. Scaled video model training to hundreds of GPUs.

EXPERIENCE

Roblox California
Principal Software Engineer *2021-Present*

VideoGen Action-conditioned real-time video and video-to-video generation

- Built the team's first video data pipeline: ingestion, filtering, deduplication, captioning, latent extraction, and automated quality scoring.
- Scaled training to hundreds of GPUs on a shared PyTorch stack with FSDP, tensor parallelism, and sequence parallelism.
- Trained action-conditioned real-time video and video-to-video models for controllable generation and long-horizon rollouts.
- Optimized GPU kernels for video generation workloads.
- Designed bucketed eval sets and model-based scoring that isolate temporal and long-horizon failure modes, and drove the next round of model and data work off those results.

MeshGen (Roblox Cube) Discrete representations and autoregressive mesh generation

- Advanced 3D representation learning by doubling the mesh tokenizer from 512 to 1024 tokens per shape and co-designing the training recipe and data pipeline, closing most of the quality gap between discrete VQ and continuous KL-VAE representations.
- Took Cube MeshGPT from research experiments to production for text- and bounding-box-conditioned 3D asset generation, spanning model training, evaluation, and inference.

Production ML Real-time avatars and safety

- Shipped models for voice moderation, audio-driven facial animation, lip sync, face tracking, and personalized avatars, running on client devices and in the cloud.

Loomai - *Deep Learning Consultant, Part-time* *Remote / Beijing, 2019-2021*

Built compact on-device models for audio-driven facial animation, activity detection, and speaker verification.

Institute of Automation, Chinese Academy of Sciences - *Postdoctoral Researcher* *Beijing, 2019-2020*

Built reinforcement-learning systems for games; mentored work on detection, tracking, and pose estimation.

Earlier research internships. Microsoft Research Asia (2017), representation learning; Intel (2015-2016), visual recognition.

TECHNICAL STRENGTHS

Modeling: 3D representation learning, mesh tokenization, autoregressive mesh generation (MeshGPT), VQ and KL-VAE; action-conditioned video, video-to-video, world models, diffusion and flow matching

Training: PyTorch, FSDP, tensor and sequence parallelism, training on hundreds of GPUs, distributed evaluation, inference optimization

Systems: Python, C/C++, CUDA, Linux, Ray, data pipelines, client and cloud inference

OPEN SOURCE AND PUBLICATIONS

Roblox Cube. Co-developed and open-sourced a foundation model for 3D shape generation and understanding; 1.2K+ stars.

ML research implementations. Deformable ConvNets v2 in PyTorch with custom CUDA kernels, used by CenterNet and other projects; Mask R-CNN and Faster R-CNN reimplementations; 5K+ stars and 1.8K+ forks combined.

2025 - Cube: A Roblox View of 3D Intelligence. Roblox Foundation AI Team (incl. Charles Shang). arXiv:2503.15475.

2025 - Efficient Autoregressive Shape Generation via Octree-Based Adaptive Tokenization. Deng, Liu, Zhu, Sun, C. Shang, et al. arXiv:2504.02817.

2024 - Diffusion Synthesizer for Efficient Multilingual Speech-to-Speech Translation. Hirschkind, Yu, Nandwana, et al. (incl. Charles Shang). Interspeech 2024.

EDUCATION

Tsinghua University - *Ph.D., Computer Vision and Machine Learning; advisor: Prof. Haizhou Ai* *Beijing, 2013-2019*

Northwestern Polytechnical University - *B.E., Computer Science; National Scholarship* *Xi'an, 2008-2013*